Civics Group	Index Number	Name (use BLOCK LETTERS)

H2

ST. ANDREW'S JUNIOR COLLEGE
2018 JC2 PRELIM

H2 BIOLOGY **9744/2**

Paper 2

Monday 10th September 2018 2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagram, graph or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)
Answer **all** questions.
Write your answers in the spaces provided on the question paper.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiners' Use	
Section A	
1	/13
2	/13
3	/14
4	/14
5	/16
6	/10
7	/9
8	/5
9	/6
Total	**/100**

This document consists of **22** printed pages.

[Turn over

QUESTION 1

Fig 1.1 shows the structure of a G-protein-linked receptor (GPLR) from a cross-section of the plasma membrane.

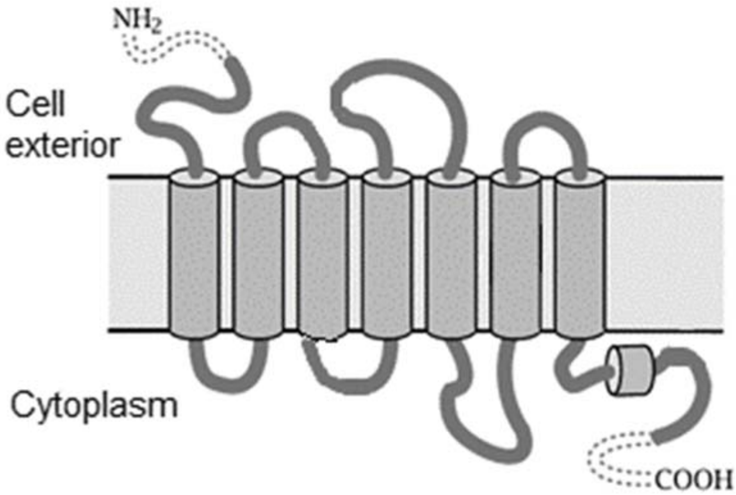

Fig 1.1

(a) With reference to **Fig 1.1**, describe the significance of R groups to the structure and function of a GPLR.

…..…………………………………………………………………………………….. [5]

1. Ref. seven **transmembrane domains/segments**, consisting of **amino acids with hydrophobic R groups**
2. that have <u>hydrophobic interactions</u> with **hydrophobic core** of plasma membrane;
3. **hydrophilic/polar** R groups of amino acids interacting with hydrophilic **phosphate heads** / **aqueous** medium;
4. ref. hydrogen bonds / ionic bonds (Note: hydrophilic interactions are vague)

5. ref. **extracellular** segment/domain/binding site which binds to **signal molecules /ligand**
6. ref. **intracellular/cytoplasmic** segment/domain/binding site which binds to <u>G-protein</u>
7. Ref. R groups of amino acids at binding site forming temporary interactions / providing complementary 3D conformation for signal molecules/G protein to bind

(b) The GPLRs make up the largest family of cell surface receptors. Outline the route taken by the GPLR after its synthesis to its final location in the plasma membrane.

…..……………………………………………………………………………………….. [4]

1. Newly synthesised polypeptide enters the rough endoplasmic reticulum (rER**)** and **folds into its native / 3-dimensional conformation**
2. Protein undergoes **chemical / post-translational modification** (where short carbohydrate chains are added to these proteins (glycosylation))
3. ref. GPLR being packaged into **transport vesicles** which **buds off the rER** and **fuse with** cis **face of the Golgi body**
4. where **further chemical / post-translational modification** of the protein occurs.
5. Modified proteins packaged into **secretory vesicles** which **bud off the** trans **face** of the Golgi body.
6. Secretory vesicles move to and **fuse with the cell surface membrane** / plasma membrane, GPLR embedded within plasma membrane;
7. Ref. movement of vesicles on cytoskeleton / microtubules involving ATP hydrolysis

GPLRs are found to be closely associated with a type of G protein called K-Ras in the cell signaling pathways.

Fig. 1.2 is a simplified diagram showing the normal roles of GPLR and K-Ras in the RAS/MARK signaling pathway.

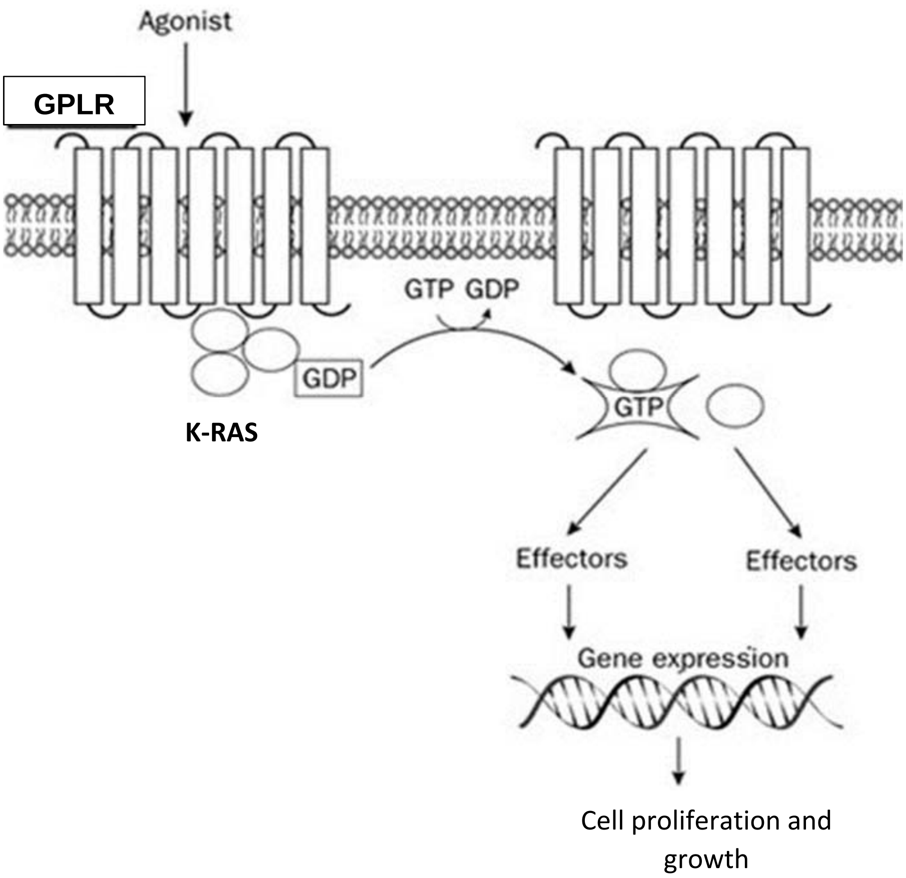

Fig. 1.2

(c) Assuming that the effectors (in **Fig. 1.2**) in the transduction pathways function normally, explain how a mutation can lead to the formation of tumours in cancer.
…..…………………………………………………………………………………….. [4]

1. (gain of function) Mutation in the K-Ras gene
2. ref. K-Ras protein functions as a GTPase enzyme;
3. Mutated K-Ras protein unable to hydrolyse/convert GTP to GDP / mutated K-Ras continuously/constitutively bound to GTP;
   ref. mutated K-Ras protein constitutively activated/switched on
4. activating/switching on downstream **transduction** pathways / ref. expression of genes; that lead to continuous cell growth and division;
5. Ref. accumulations of other mutations / 1 example e.g. tumour suppressor genes / other proto-oncogene mutations

Or

1. Mutation in the GPLR gene
2. Mutated GPLR unable to hydrolyse ligand / ligand continuously bound to mutated GPLR / doesn't require ligand for activation;
3. GPLR continuously activated to (expose K-Ras binding domain to) bind to G-protein;
   K-Ras protein continuously activated/switched on
4. activating/switching on downstream **transduction** pathways / ref. expression of genes that lead to continuous **cell growth and division;**
5. Ref. accumulations of other mutations + 1 example e.g. tumour suppressor genes / other proto-oncogene mutations

QUESTION 2

Fig. 2.1 shows Process X in an eukaryotic cell which produces ribosomal RNA (rRNA).

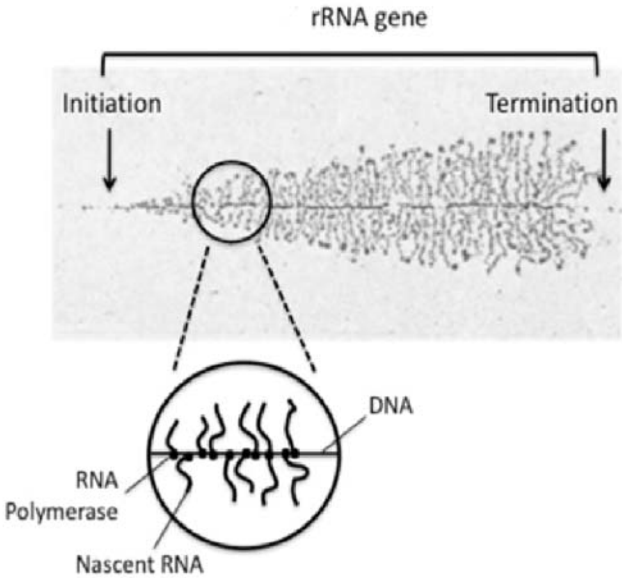

Fig. 2.1

(a)(i) Name the Process X occurring in **Fig. 2.1**.
……………………………………………………………………………………………….[1]

1. Transcription

(ii) List one molecule **not mentioned in Fig. 2.1** that is required for Process X.
……………………………………………………………………………………………….[1]

1. General Transcription factor (Reject: Specific transcription factor due to it not a real requirement for transcription)
   / ribonucleotides
   / transcription initiation factors;

(iii) Describe how RNA polymerase is able to recognise and bind to the promoter on DNA and not to other DNA regions.
……………………………………………………………………………………………….[2]

1. Ref. RNA polymerase has a domain complementary in shape to general transcription factors which bind to TATA box of promoter

2. General Transcription factors contain a **DNA-binding domain** [Reject: active site] which recognize and bind to specific DNA sequence in the **promoter** ;
3. Ref. **Nucleotide sequence** / length / major and minor grooves of promoter offers a **complementary shape** to DNA-binding domain of RNA polymerase ;

[Reject: complementary base pairing]

(iv) Explain for the observed pattern of Process X in **Fig. 2.1**.
…………………………………………………………………………………………………….[2]

1. **Shorter** RNA transcripts seen at the **beginning** of the DNA template strand, which get **longer** till the **end** of the transcription unit, (where the transcripts detach from the DNA template after transcription termination)
2. Due to simultaneous transcription of rRNA gene **by multiple RNA polymerases**, (causing RNA transcripts to extend perpendicularly from DNA template strand) ;

(v) State the roles of rRNA in protein synthesis.

…………………………………………………………………………………………………….[2]

1. The rRNA in ribosomes holds the tRNA and mRNA together in **close proximity**, via **complementary base pairing / hydrogen bonds**
2. positions the new amino acid for addition to the <u>carboxyl end</u> of the growing polypeptide
3. rRNA <u>peptidyl transferase</u> activity catalyzes formation of a <u>peptide bond</u> between the new amino acid and the polypeptide chain
4. Ref. rRNA associate with proteins to form ribosomal subunits / ribosomes (which synthesizes proteins)

(b) During protein synthesis in cells of an embryo, all tRNA molecules with UAC anticodon sequence, are observed to be bound to arginine amino acid instead of methionine.

(i) Suggest how these tRNA molecules attached with the wrong amino acid might arise.
…………………………………………………………………………………………………….[2]

1. Ref. possible mutation in the **gene** sequence for the aminoacyl tRNA synthetases,
2. resulting in **altered 3D conformation** of active site
   which is **complementary** (in shape) to the amino acid arginine and the corresponding tRNA with anticodon UAC

(ii) Suggest and explain the effect of this wrong pairing of amino acid to tRNA on the embryo.
…………………………………………………………………………………………………….[3]

1. ref. altered **primary sequence** of polypeptides (all methionine replaced by arginine) and folding of polypeptides to **tertiary structure** / 3D conformation is affected;
2. ref. **non-functional proteins** made in cells
3. ref. possible disruption of metabolic processes in the **cell** / cells might die easily, **embryo cannot further develop** into a fetus

[Total: 13m]

QUESTION 3

In a dihybrid inheritance, gene B/b codes for flower colour while gene H/h codes for leaf shape of a plant.

The F1 progeny of a pure-bred plant with red flowers and oval leaves, and another pure-bred plant with yellow flowers and fan-shaped leaves, have red flowers and fan-shaped leaves.

F1 plants then undergo a test cross.

(a) Predict the expected phenotypic ratio in the F2 progeny.
…………………………………………………………………………………………………[1]

1. 1 Red flower, fan-shaped leaf : 1 Red flower, oval leaf : 1 Yellow flower, fan-shaped leaf: 1 Yellow flower, oval leaf
   [Reject: 1:1:1:1 with no phenotypes]

(b) Explain how different characteristics can be inherited independently in dihybrid inheritance.
…………………………………………………………………………………………………[2]

1. The **2 genes** (encoding for flower colour and leaf shape) are found on **different** pairs of homologous **chromosomes** / are **unlinked** genes ;
2. Ref. **independent assortment** occurs (Reject: random assortment)
   / allows for random **arrangement** of the alleles of one gene pair at **metaphase plate** during metaphase I and II is independent of the alleles of the other gene pair ;
   Subsequent segregation of alleles of one gene pair is independent of the alleles of the other gene pair during anaphase I and II;
   (allows the production of gametes with different combinations of alleles)

(c) Using the symbols for the alleles stated above, draw a genetic diagram to show the expected phenotypic ratios for the offspring of the test cross if inheritance is Mendelian.

……………………………………………………………………………………………….[3]

F_1 phenotypes:	Red flower, Fan-shaped leaf	x	Yellow flower, Oval leaf
F_1 genotype:	BbHh		bbhh
F_1 gametes	BH Bh bH bh		bh

F_2 genotypes:
<u>Punnett square:</u>

	BH	Bh	bH	bh
bh	BbHh (Red flower, Fan-shaped leaf)	Bbhh (Red flower, Oval leaf)	bbHh (Yellow flower, Fan-shaped leaf)	bbhh (Yellow flower, oval leaf)

F_2 / Progeny phenotypes:	Red flower, Fan-shaped leaf	Red flower, Oval leaf	Yellow flower, Fan-shaped leaf	Yellow flower, oval leaf
F_2 / Progeny phenotypic ratio:	1	1	1	1

Mark scheme:
1. F_1 phenotype and genotypes
2. Parental gametes – (Gametes **must** be circled)
3. F_2 genotypes correspond to phenotypes

(d) You will now proceed to do a chi-square statistical test for the above cross.

(i) State the objective of performing a chi-squared statistical test.

…………………………………………………………………………………………………[1]

1. To test if there is **significant difference** between **observed and expected** numbers / results (Reject: ratio)

The number and phenotypes of F2 plants are listed below:

F2 phenotypic classes	Observed numbers
Red flower, fan-shaped leaf	85
Red flower, oval leaf	70
Yellow flower, fan-shaped leaf	89
Yellow flower, oval leaf	78

Formula for χ^2 calculation

$$\chi^2 = \Sigma \frac{(O-E)^2}{E} \qquad v = c - 1$$

where Σ = 'sum of…' O = observed 'value'
v = degrees of freedom E = expected 'value'
c = number of classes

(ii) Calculate the chi-square value.

…………………………………………………………………………………………………[2]

1. 2.60 (2 dp according to chi-square table);
2. 1 mark for clear working

Table 3.1

degrees of freedom	probability P				
	0.50	0.10	0.05	0.01	0.001
1	0.46	2.71	3.84	6.64	10.83
2	1.39	4.61	5.99	9.21	13.82
3	2.37	6.25	7.82	11.35	16.27
4	3.36	7.78	9.49	13.28	18.47

(iii) Using Table 3.1 and the calculated chi-square value, find the probability that observed and expected results differ by chance.

……………………………………………………………………………………………………….[1]

1. Probability is between 0.10 and 0.50 (Reject: between 0.50 and 0.10) / between 10% and 50%

(iv) State the conclusions for this test.

……………………………………………………………………………………………………….[2]

1. There is **no significant difference** between observed and expected values/results, **any difference is due to chance**
2. The Mendelian ratio of 1:1:1:1 is correct

(v) Plants are a good choice of experimental organisms for carrying out such crosses and for performing statistical tests.

Compared to plants, humans are less ideal and it is usually more difficult to arrive at reliable conclusions for observations involving humans. Suggest why.

……………………………………………………………………………………………………….[2]

1. Humans produce **limited offspring**. (This makes statistical tests difficult)
2. Humans have a long life span and **some traits only appear at a later stage in life**

[Total: 14m]

QUESTION 4

(a) In many ant species, polymorphism exists in the form of worker and queen ants. Some distinct differences between workers and queens are that workers are much smaller and cannot reproduce. Strict caste roles are also observed: queens lay eggs and workers take care of all other work, including offspring.

In a new study to identify the cause for these behavioural and physical differences, Rockefeller scientists report that a gene coding for an insulin-like peptide, ILP2, is instrumental in promoting and suppressing reproduction.

ILP2 is the ant version of insulin and, like human insulin, regulates metabolism by cellular uptake of glucose.
The table below summarises their results:

Expression of ILP2	**Type of ants**
High	Reproducing
Low	Non-reproducing

(i) Suggest a link between glucose uptake and reproduction.

……………………………………………………………………………………………. [1]

1. ref. link between **nutritional state/ATP synthesis** and production of viable/fertile/healthy offspring;

(ii) The presence of larvae causes the activation of ovaries in worker ants. It was suggested that larvae release pheromones that control the expression of the ILP2 gene.

Explain how pheromones could have played a role at the transcriptional level of ILP2 expression in determining the caste roles of ants.

……………………………………………………………………………………………. [4]

1. Pheromones act as activator/specific transcription factor; (Reject: general transcription factor due to the word "control" of expression)
2. **Binding** of activator / specific transcription factors to <u>enhancer</u>
3. Faciltates the <u>efficient positioning of RNA polymerase</u> at promoter / stabilizes the <u>transcription initiation complex</u> (to increase the rate of transcription)
4. Increased expression of ILP2 gene that lead to **activation of ovaries** => **queen ants**

(iii) The expression of the ILP2 gene can be further controlled even after translation of the ILP2 mRNA. Describe 2 ways to control ILP2 expression at this level.

…………………………………………………………………………………………..[2]

1. Transport of ILP2 protein to target destinations;
2. Chemical modification / attaching other biochemical functional groups / any 1 example i.e. phosphorylation, acetylation, glycosylation and the formation of disulfide bonds;
3. <u>Ubiquitin</u> are covalently attached to ILP2 proteins which are **degraded** by proteasomes;
4. **Proteolysis / hydrolytic processing/cleavage** of polypeptide into smaller, functional protein molecules;

(b) Double-stranded DNA of the ILP2 gene is denatured and allowed to hybridise through complementary base pairing with mature ILP2 mRNA isolated from queen ant cells. This is shown in **Fig. 4.1**.

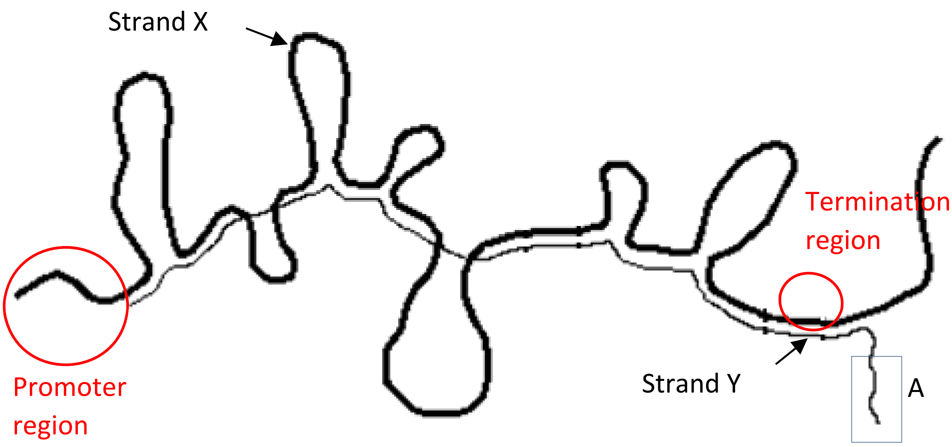

Fig 4.1

(i) State the identity of Strand X.

....[1]

1. Template DNA strand / single-stranded DNA (of ILP2 gene)
REJECT: DNA (which is double-stranded)

(ii) Explain your answer in (i) using evidence from Fig. 4.1.

....[2]

1. Strand X longer
2. due to introns / promoter / terminator

OR

1. Strand Y shorter
2. due to introns being excised (to form the mature mRNA) / as promoter / terminator not transcribed

OR

1. Strand X contains **looped** portions that consist of **introns**
2. No complementary sequence on Strand Y (mature mRNA) for complementary base pairing with non-coding sequences in Strand X

(iii) Circle the locations of the promoter and termination regions of the ILP2 gene on **Fig. 4.1**. Label the regions accordingly. Ans in RED. [2]

(iv) Suggest the identity of the unpaired segment (labelled A) on strand Y. Explain your answer.

....[2]

1. Poly (A) tail
2. Poly (A) tail is added to strand Y post-transcriptionally /Template DNA does not code for the poly (A) tail

[Total: 14m]

QUESTION 5
(a) The following extract was summarised from an article published in Proceedings of the National Academy of Sciences journal.

The article proposes a method to overcome HIV pandemic using genetically modified rice.

Summary points of article:

- Scientists from the US, UK, and Spain have developed a new strain of genetically modified rice to manage HIV symptoms in countries where traditional medicines can be hard to access.
- The rice seeds produce three proteins – the antibody 2G12, and the lectins griffithsin and cyanovirin-N – which preliminary *in vitro* tests show bind to gp120 and *neutralize* HIV.
- These seeds can be ground up to form a paste that can then be applied as a topical cream, which counterbalances the virus in the exact same way as the anti-retroviral medication.
- There remains a few *hurdles* researchers will have to jump before the rice becomes widely available.

(i) Define the term "neutralise" in this context and explain how this process can halt the reproductive life cycle of HIV.
………………………………………………………………………………………………[3]

1. Ref. antibody 2G12, / lectins griffithsin / cyanovirin-N **bind to/surround** gp120 of HIV
2. Gp120 **cannot** recognise and bind/adsorb to CD4 receptor of T helper cells/macrophages;
3. Ref. fusion between viral envelope and host cell plasma membrane **cannot** occur

(ii) Suggest one possible hurdle that researchers need to overcome for this treatment using GM rice.
………………………………………………………………………………………………[1]

1 Ref. unknown long term health consequences
2 Ref. acceptance to tampering of nature
3 Ref. GM rice disrupting eco-system
4 AVP e.g. rapid changing antigenic surface of gp120
[Any 1]

(b) Fig 5.1 shows a lentivirus, which can bind to cells lining the airways of the lungs. The lentivirus is a form of **retrovirus**. The **general structure** of this virus is similar to that of HIV.

The lentivirus is commonly used as a vector (vehicle to transport external copies of RNA coding for specific proteins into cells).

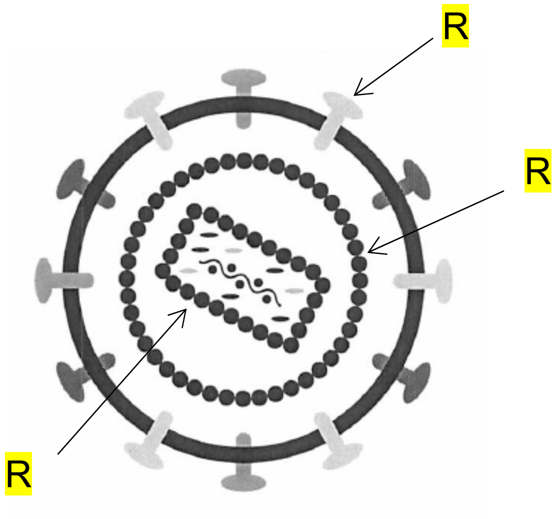

Fig. 5.1

(i) Using the letter **R**, label **Fig 5.1** to identify a feature that is protein in nature.
……………………………………………………………………………………………[1]

1. Glycoprotein
2. Matrix
3. Capsid
4. Viral enzyme(s)

[Any 1]

(ii) Explain why external copies of RNA intended to be introduced into cells cannot pass through the membrane of cells directly.
……………………………………………………………………………………………[2]

1. RNA is <u>charged</u> (due to phosphate backbone) (Reject: large size)
2. Cannot pass through hydrophobic core of fatty acid tail / phospholipid bilayer.

(iii) With reference to your knowledge on retroviruses, explain how **long-term** expression of an inserted RNA is brought about following infection of host cells with the lentiviral vector.
……………………………………………………………………………………………[3]

1. Inserted RNA (and viral RNA genome) is **reverse transcribed** into cDNA using viral <u>reverse transcriptase</u>
2. the resulting cDNA is then **integrated** into host cell DNA using viral <u>integrase</u>
3. cDNA (of inserted RNA) is **expressed** using host cell machinery e<u>.g. RNA polymerase and host ribsomes</u>

(c) Plasmids are small, self-replicating circles of DNA.

During cloning, a target gene can be inserted into a plasmid, making the plasmid a **vector** that transports the target gene into cells.

It is hoped that the plasmid and the target gene gets replicated to multiple copies in the cell.

Based on its features, suggest why plasmids are good choices of cloning vectors.
……………………………………………………………………………………………[1]

1 Small **size** plasmid allows **insertion of target genes of larger sizes / ease of being taken up by cells;**

2 Ref. **Origin of replication** allows for plasmid to **replicate independently**, (results in multiple copies of the plasmid and inserted foreign gene within one bacterium)

3 **Double stranded** DNA nature similar to the target gene allows for **integration**

[Any 1]

Reject: points on conjugation as question is on cloning in the lab.

(d) The F plasmid is a crucial plasmid found naturally in bacterial cells. It plays an integral role in the transfer of genes from one bacterial cell to another through processes such as conjugation.

Fig. 5.2 shows the process of conjugation.

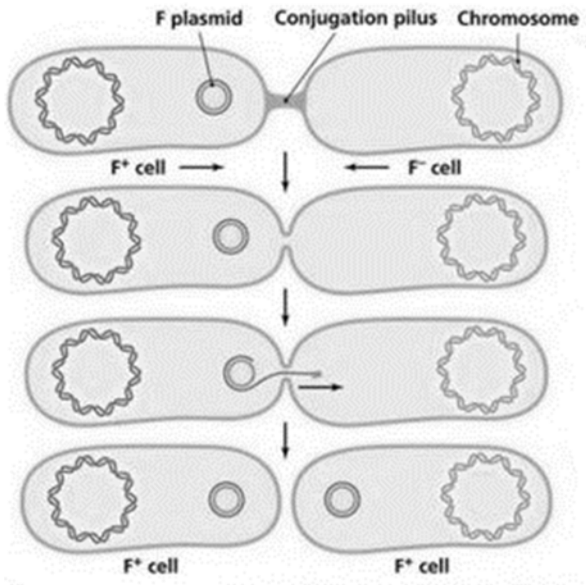

Fig. 5.2

During conjugation, the transferred F plasmid strand undergoes DNA replication similar to the synthesis of Okazaki fragments.

Explain why.

…………………………………………………………………………………………………….[2]

1. <u>DNA polymerase</u> can only add new nucleotides to the **available** <u>3'-OH</u> end of a pre-existing polynucleotide chain;;
2. DNA strands are antiparallel
   /DNA strands run in the 5'→ 3' and 3'→ 5' directions ;;

(e) Scientists isolated one of the *lac permease* gene from the lac operon and proceeded to amplify the gene using an automated process of Polymerase Chain Reaction (PCR).

Fig. 5.3 shows the temperatures used at the different stages during PCR.

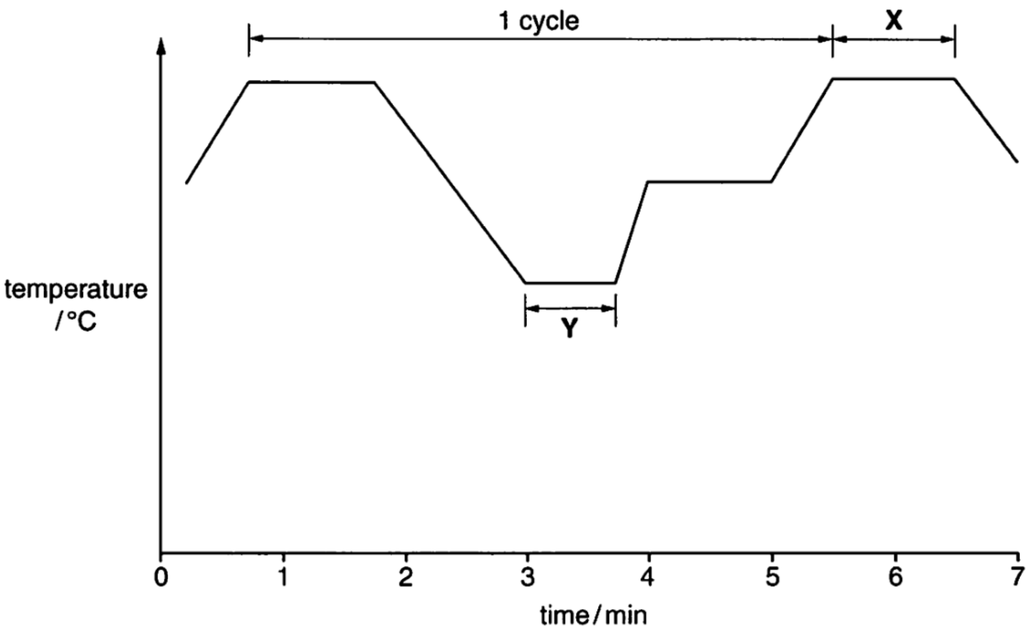

Fig. 5.3

(i) Describe what happens at stage Y.

…………………………………………………………………………………………………[2]

1. Annealing stage at temperatures between 50 – 65^{0}C ;
2. **Forward and reverse primers** bind to complementary sequences **flanking** the target sequence to be amplified at the 3' ends of single (DNA) strands ;

(ii) If stage X of PCR process persisted for more than two hours, suggest what you will expect to find in the PCR mixture during this period.

…………………………………………………………………………………………………[1]

1. Free/unused DNA nucleotides / DNA primers ;
2. Some Taq DNA polymerases are denatured, (hence they are unable to bind to and elongate DNA) ;
3. **Single-stranded** DNA templates (with no new daughter molecules)

[Total: 16m]

QUESTION 6

(a) The concentration of carbon dioxide in a sample of air was found to be 280 ppm (parts per million).

An experiment was designed to measure the concentration of carbon dioxide in the air after it had flowed over the leaves of a green plant. Measurements were taken at a range of light intensities.

The following results were obtained:

Light intensity (% of full sunlight)	**Concentration of carbon dioxide in air after flowing over leaves (ppm)**
75	253
50	252
25	254
10	280

(i) Predict the concentration of carbon dioxide in the air after flowing over the leaves when the plant was placed in the dark.

…..…………………………………………………………………………………….. [1]

1. 280ppm or more;

(ii) Explain the experimental results obtained in relation to light intensity being a limiting factor of photosynthesis.

…..…………………………………………………………………………………….. [3]

1. ref. light intensity being a limiting factor of photosynthesis at values **lower than 25% or 50%**;
2. As light intensity increases from 10% to 25% or 50%, concentration of carbon dioxide in air after flowing over leaves **decreases from 280ppm to 254ppm (or 252ppm for 50%**)**;**
3. At higher light intensities tested (i.e. 25% or 50% and above), the rate of photosynthesis is affected by **factors other than light**;
   / light intensity is no longer a limiting factor;
4. Concentration of carbon dioxide in air after flowing over leaves **remains constant around 252ppm-254ppm**, due to the rate of photosynthesis being constant;
5. Ref. light intensity leads to higher electron flow down ETC and higher production of **ATP and NADPH**, leading to **higher rate of Calvin cycle** where CO2 fixation occurs

(b) Discuss the validity of the statement "The synthesis of ATP during photophosphorylation depends on the **transport of protons** across membranes."
…..…………………………………………………………………………………….. [4]

1. Statement is valid;
2. Energy released by the transfer of electrons down the ETC used to <u>**pump**</u> H^+ ions / protons **from stroma into thylakoid space**;
3. Higher concentration of H^+ ions in thylakoid space than in stroma / proton gradient established across thylakoid membrane;
4. **Diffusion** / **facilitated diffusion** of H^+ ions / protons (down concentration gradient) through ATP synthase
5. Potential energy coupled to ATP synthase to produce ATP from ADP and Pi;

(c) $NADP^+$ is an important electron carrier found in plants, but its level decreases sharply in the day. Suggest the significance of the decrease of $NADP^+$ during the day.
…..…………………………………………………………………………………….. [2]

(ref. light-dependent reactions during day
/ photoexcited electrons (from PS I and II)being passed down ETC)

1. $NADP^+$ acts as the **final electron acceptor**, forming <u>NADPH</u>;
   / $NADP^+$ **accepts electrons and H^+ ions** to form <u>NADPH</u>;
2. NADPH (and ATP) used in **Calvin cycle** / light independent reactions to reduce PGA to PGAL **to synthesis of glucose;**

[**Total: 10m]**

QUESTION 7

Ants and bees – which by all appearances seem so different – are found to be close relatives (refer to **Fig. 7.1**) in a new study which used cutting-edge DNA sequencing techniques to elucidate the taxonomic relationships among the different families of wasps, bees and ants.

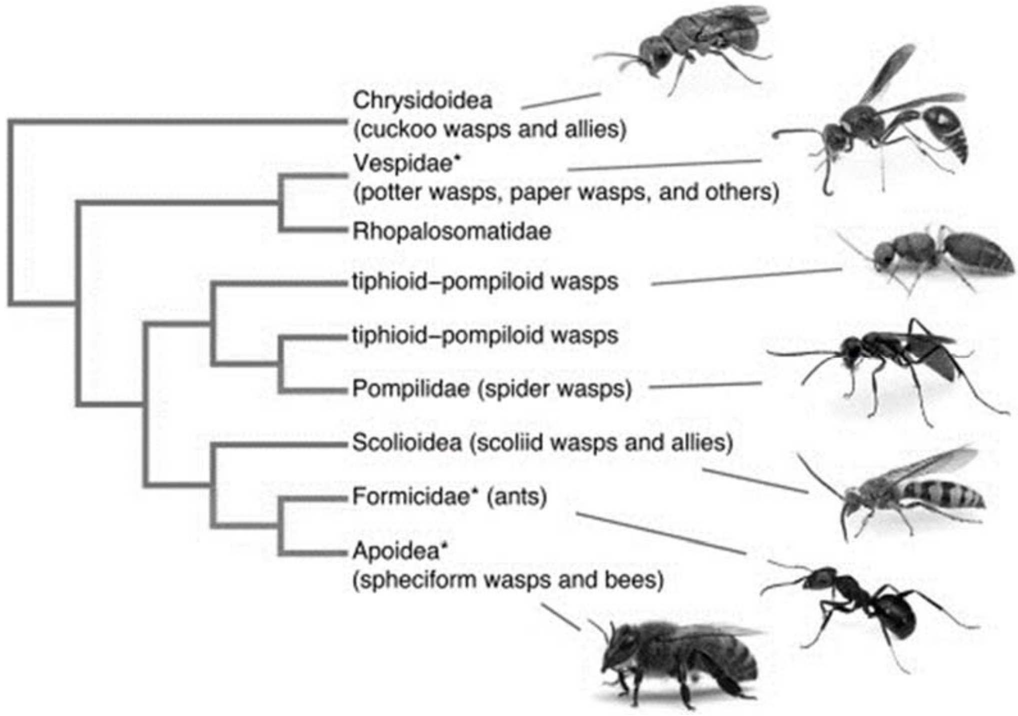

Fig. 7.1

(a) State **one** advantage of using DNA sequences to elucidate the evolutionary relationship between ants and wasps.

…..……………………………………………………………………………………….. [1]

1. Quantifiable / open to statistical analysis;
2. Unambiguous and **objective**;
3. Not affected by convergent evolution;
4. Based strictly on heritable material;
5. Greater number of organisms can be compared;
6. Greater number of characters can be compared;
7. Ref. abundance in DNA samples

(b) Discuss how molecular homology was used in this study to derive the phylogram in **Fig. 7.1**.

…..……………………………………………………………………………………….. [2]

1 Comparison of DNA sequences of a **common gene** from the different families;

2 High homology indicates that they are more closely related / share a more recent common ancestor (vice versa);

(c) Explain how biogeography can help to support the evolutionary deductions in this case study.

…..…………………………………………………………………………………….. [2]

1. ref. biogeography being the study of the geographic distribution of species;
2. Species of bees and ants that are **closely related /recently evolved from a common ancestor** are **found close together**,;
   /species that are not closely related are found in environments far away due to geographical barriers;

(d) The same study also suggested that when the early bees switched from preying on other insects to pollen feeding, the number of bee species exploded compared with their hunting wasp sister group. The switch from predatory behavior in hunting wasps to pollen feeding in bees also saw a corresponding increase in diversification of the mouthparts and tongue lengths of the bees.

Suggest and explain how the increase in number of bee species is made possible with the switch to pollen feeding, as compared to speciation in hunting wasps.

…..…………………………………………………………………………………….. [4]

1. More types of flowers available as compared to types of insects/prey;
2. **<u>Variation</u>** in the mouthparts and tongue length of early bees due to **mutation**;
3. Different <u>selection pressures</u> + different types of food available;
4. Individuals with a <u>selective advantage</u> in the particular environment **survive** till **reproductive** age; and pass on their <u>alleles</u> to their offspring; (Reject: traits)
5. ref. <u>reproductive</u> barriers/isolation / preventing <u>gene flow</u> between different populations of bees (leading to each population accumulating own genetic differences);
6. ref. **long period of time** before different species evolved;
7. ref. adaptive radiation / divergent evolution;

[Total: 9m]

QUESTION 8

(a) Explain how the structure of antibodies cause pathogens to be destroyed by macrophages.

……………………………………………………………………………………………………[3]

1. **Variable / antigen-binding sites** of antibodies allow for (recognition and) **binding to pathogens**
2. **Constant region** (of heavy chains) allow (recognition and) **binding** (of antibodies) **to** (Fc) <u>receptors</u> on **macrophages**
3. Ref. **complementary shape recognition** between antigen-binding site and pathogen / between constant region of antibody with macrophages (followed by subsequent phagocytosis of pathogens by macrophages)

(b) DNA coding for the light chain of an antibody is isolated from a **mature** B lymphocyte.

DNA coding for the light chain of an antibody is also isolated from a **developing** B lymphocyte (before maturation).

Gel electrophoresis was performed on both sources of DNA.

Fig 8.1 shows the gel diagram and the corresponding positive / negative electrode.

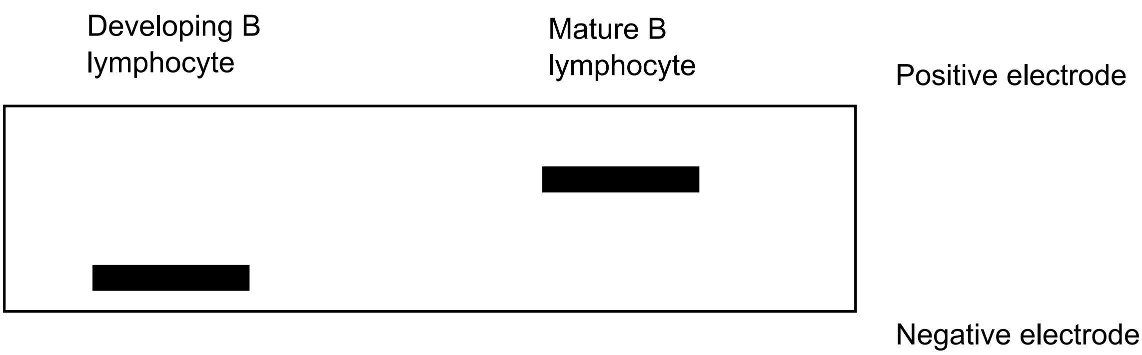

Fig. 8.1

Explain the difference in the results on the gel diagram.

……………………………………………………………………………………………………[2]

1. DNA fragment from mature B lymphocyte travels <u>faster</u> (Reject: further) than that of the developing B lymphocyte / DNA fragment from mature B lymphocyte is short**er** in length (Accept: vice versa)
2. Ref. <u>somatic recombination</u> has **already** occurred in mature B lymphocytes / certain V,J,C segments were chosen and the **rest of the segments removed** (thus, DNA is shorter) (Accept: vice versa)

[Total: 5m]

QUESTION 9

Carbon dioxide is one of the greenhouse gases that contribute to global warming and subsequently climate change.

A group of high school students decided to test whether varying temperatures would correspondingly affect mean carbon dioxide gas emission.

Table 9.1 shows the mean carbon dioxide gas emission after exposing a fixed number of mealworms to different temperatures. Carbon dioxide gas emission was measured before and after exposure to experimental temperature.

Table 9.1 showing effects of varying temperatures on mean carbon dioxide gas emission, measured in parts per million (ppm). Values represent mean ± standard deviation.

Temperature /°C	Mean Carbon dioxide gas emission /ppm	
	Before exposure to experimental temperature	**After** exposure to experimental temperature
30.0	445 ± 25	450 ± 17
40.0	450 ± 20	500 ± 30
50.0	460 ± 17	540 ± 18

(a)(i) Describe the patterns shown by the data in Table 9.1.
…………………………………………………………………………………………..……..[2]

1. The mean carbon dioxide gas emission increases as temperature increases;
2. As temperature increases from 30.0°C to 50.0°C, emission increases from 450ppm to 540ppm.
   OR
1. **Increase** in carbon dioxide gas emission **after exposure** to experimental temperatures **compared to before** exposure, gets bigger as temperature increases
2. As temperature increases from 30.0°C to 50.0°C, increase in emission increases from 5ppm to 80ppm

(ii) With reference to the mean carbon dioxide gas emission **before** exposure to 40 °C, suggest what standard deviation means.

……………………………………………….……………………………………..[2]

1. Standard deviation is the **deviation from the mean** carbon dioxide gas and determines the range of the carbon dioxide gas emission observed
2. the mean carbon dioxide gas emission ranges from 430ppm (450-20) to 470ppm (450+20)

(b) Corals are affected by rising temperatures in ocean waters. Explain how.
……………………………………………….……………………………………..[2]

1. Ref. Absorption of **more carbon dioxide which dissolves** when ocean waters get warmer; **ocean pH decreases** / ocean acidification occurs
2. Hard **corals cannot absorb calcium carbonate** they need to maintain their **skeletons**, stony skeletons that support corals will dissolve and corals destroyed / Coral polyp metabolism is affected and **corals expels the zooxanthellae**, (leaving the coral skeleton bleached), and eventual **death of corals due to lack of nutrients (provided by zooxanthellae)**

OR

1. **Photosynthesis in zooxanthellae is disrupted** at higher than usual temperatures, thus producing an excess of **products that are toxic**
2. Coral polyp metabolism is affected and **corals expels the zooxanthellae**, (leaving the coral skeleton bleached), and eventual **death of corals due to lack of nutrients (provided by zooxanthellae)**

[Total: 6m]